

Prove it, don't just install it.

How a small team builds an unsupervised bot detector with an AI agent crew, and why nothing ships until the claim behind it has survived a measurement.

SUBJECT

Bots Without Labels

DISCIPLINE

Evidence-first development

AUDIENCE

Engineers & teams adopting AI agents

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01 Why a methodology at all

Most software either works or it doesn't, and a test suite can usually tell the two apart. Our project sits in an awkward corner where that stops being true. Bots Without Labels hunts for automated traffic in server logs that nobody has labelled. No column says which rows came from a botnet. There is no answer key, and there never will be one, because the whole point of the tool is to work on logs where the truth is unknown.

Take a concrete case from our own history. Early on we built a per-entity diversity feature. The reasoning was sound: an automated actor repeats itself, so score every actor in the log by how self-similar its events are, and the monotonous ones are your bots. On a real botnet capture it lifted recall from almost nothing to 0.998, which felt like a triumph for about an hour. It also flagged more than a fifth of the log, because a nightly backup job hammering one server is every bit as monotonous as a command-and-control beacon. Nothing crashed. Every test stayed green. The feature was still unusable on its own, and the only